Summarization and graphical presentation of data(2)

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Types of Graphs

- Graphs for Qualitative data
- Graphs for Quantitative data

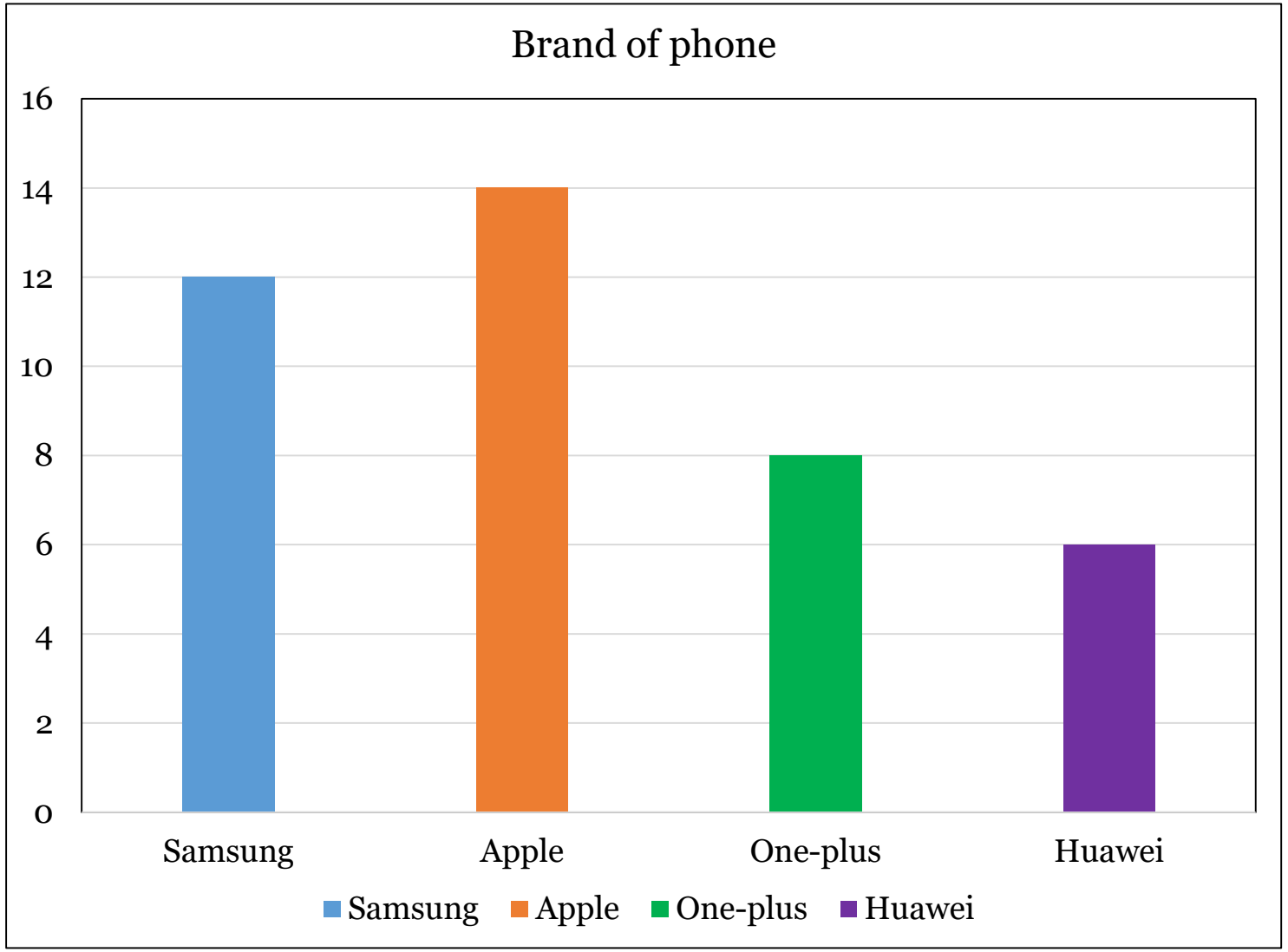
Graphs for Qualitative

- Common methods for graphically displaying qualitative data are

1. Bar Chart
2. Pie Chart

Bar Chart

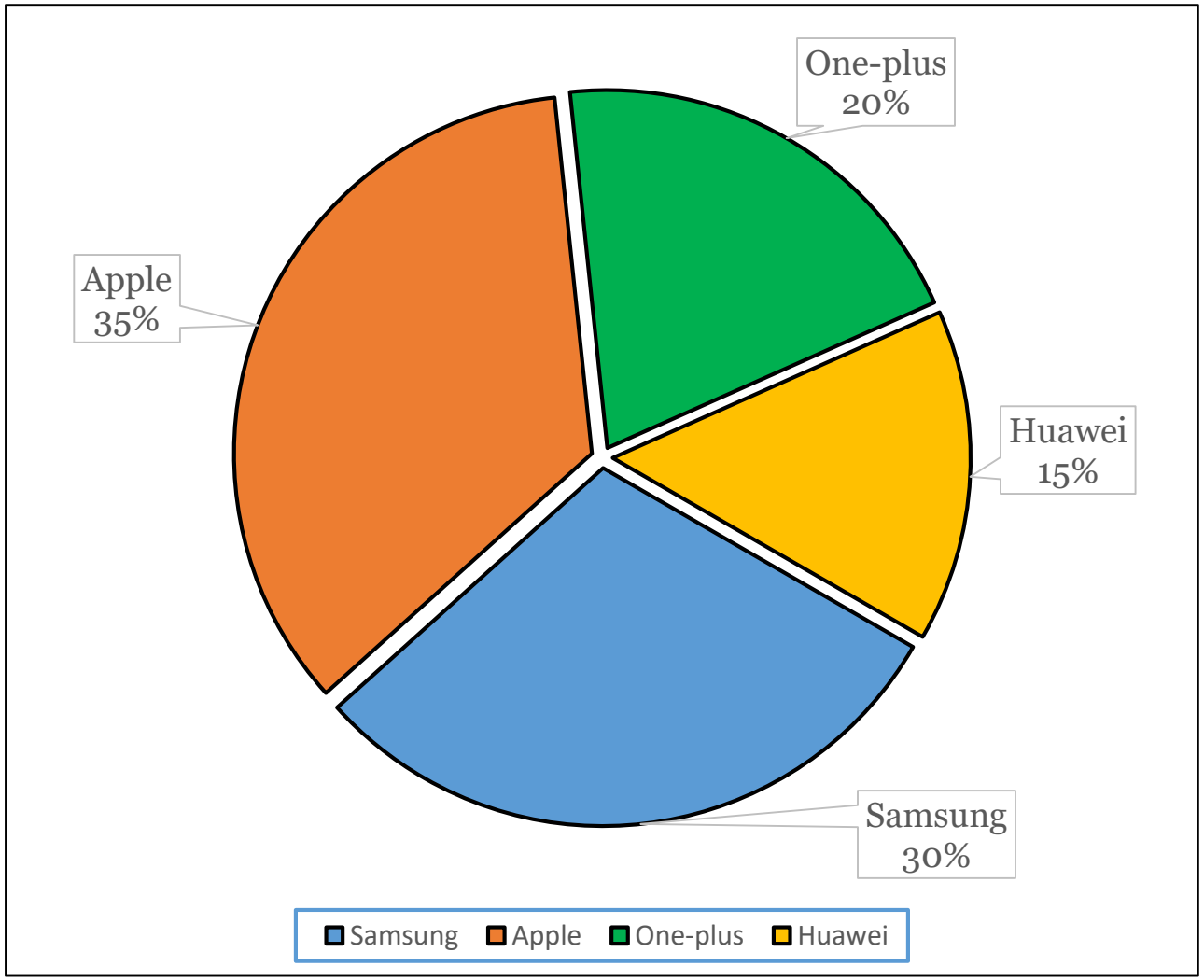
Brand	Frequency
Samsung	12
Apple	14
One-plus	8
Huawei	6
Total	40



Pie Chart

$$Slice\ size = 360 \times RF$$

Brand	Frequency	Relative frequency
Samsung	12	12/40=0.30
Apple	14	10/40=0.35
One-plus	8	0.20
Huawei	6	0.15
Total	40	1.00



Graphs for Quantitative

- Common methods for graphically displaying quantitative data are

1. Histogram
2. Frequency polygon
3. Frequency curve
4. Ogive curve
5. Stem and Leaf plot

Histogram

1. Firstly, we need to create a frame work (that is, X axis and Y axis)
2. Mark the distinct class boundary value in X-axis
3. Mark the frequencies in the Y-axis
4. Draw a rectangle above each class whose height equals to the frequency of that class.
5. Label the axis.

Histogram

- Distance of University from home (KM) for randomly selected students,

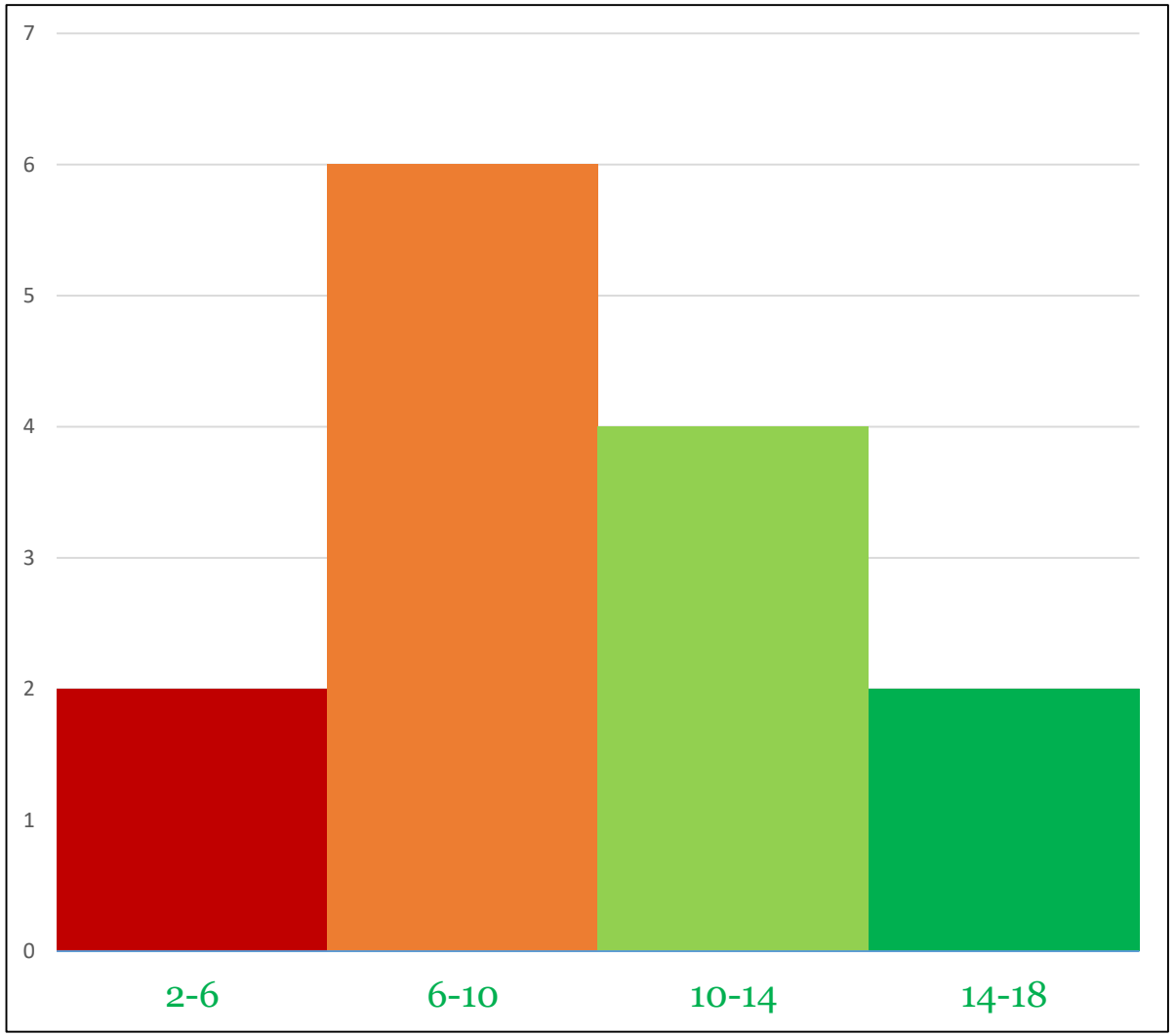
3, 8, 12, 9, 14, 6, 10, 11, 7, 5, 7, 16, 8,13

a) Construct the frequency distribution table.

b) Construct the histogram for the data.

Histogram

Class	Tally	Frequency
2-6	||	2
6-10	𝍸 |	6
10-14	||||	4
14-18	||	2



Histogram

- Age (in years) of 30 randomly selected men,

43,66,54,26,64,36,16,31,36,21,53,27,58,29,25,50,24,14,43,
34,32,26,15,22,49,55,26,11,57,15.

a) Construct the frequency distribution table.

b) Construct the histogram for the data.

Frequency Polygon

1. Firstly, we need to create a frame work (that is, X axis and Y axis)
2. Mark the distinct class mid value in X-axis
3. Mark the frequencies in the Y-axis
4. Plot a point above each mid value at the height equal to the frequency of that class
5. Label the axis.

Frequency Polygon

- Distance of University from home (KM) for randomly selected students,

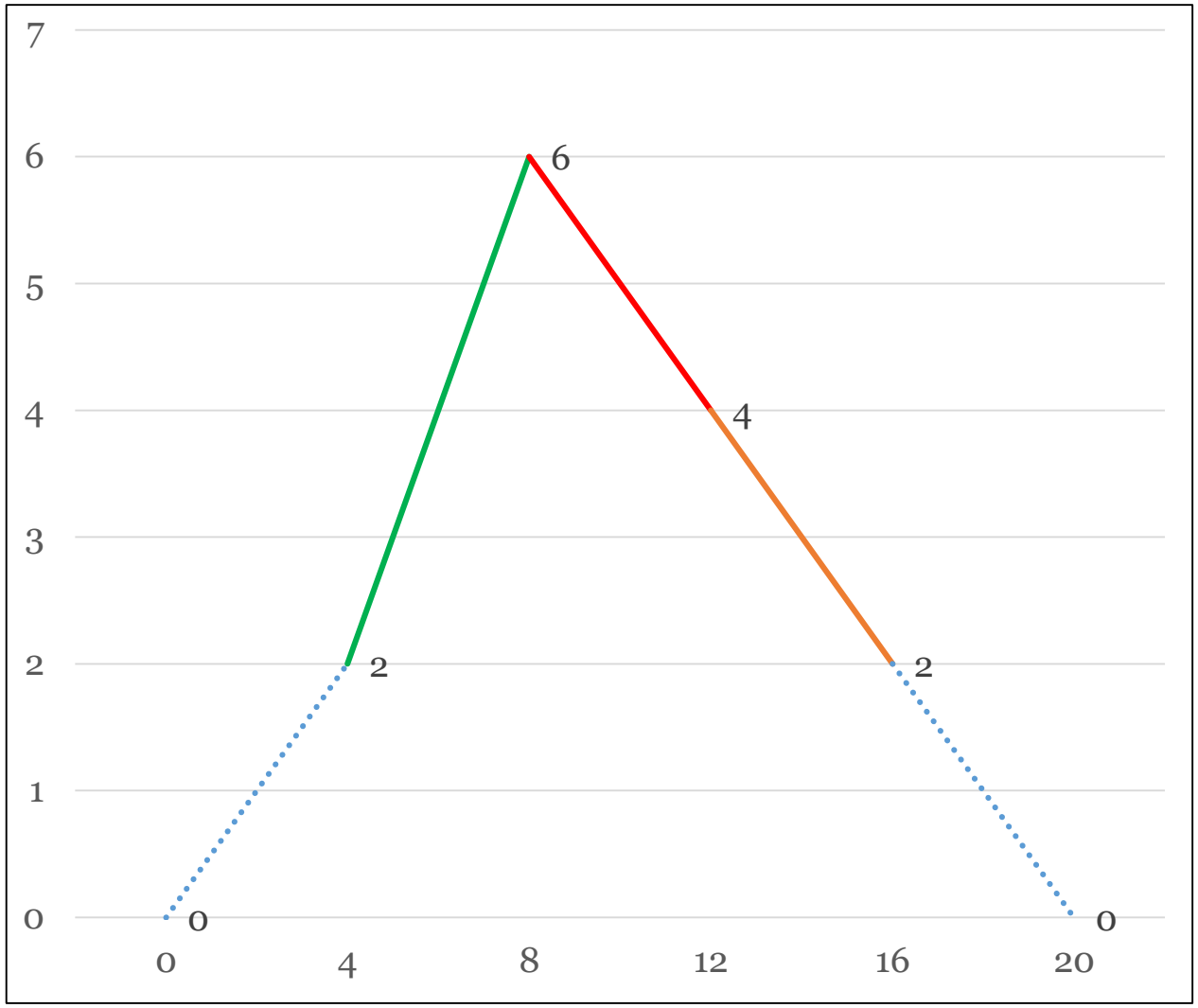
3, 8, 12, 9, 14, 6, 10, 11, 7, 5, 7, 16, 8,13

a) Construct the frequency distribution table.

b) Construct the histogram for the data.

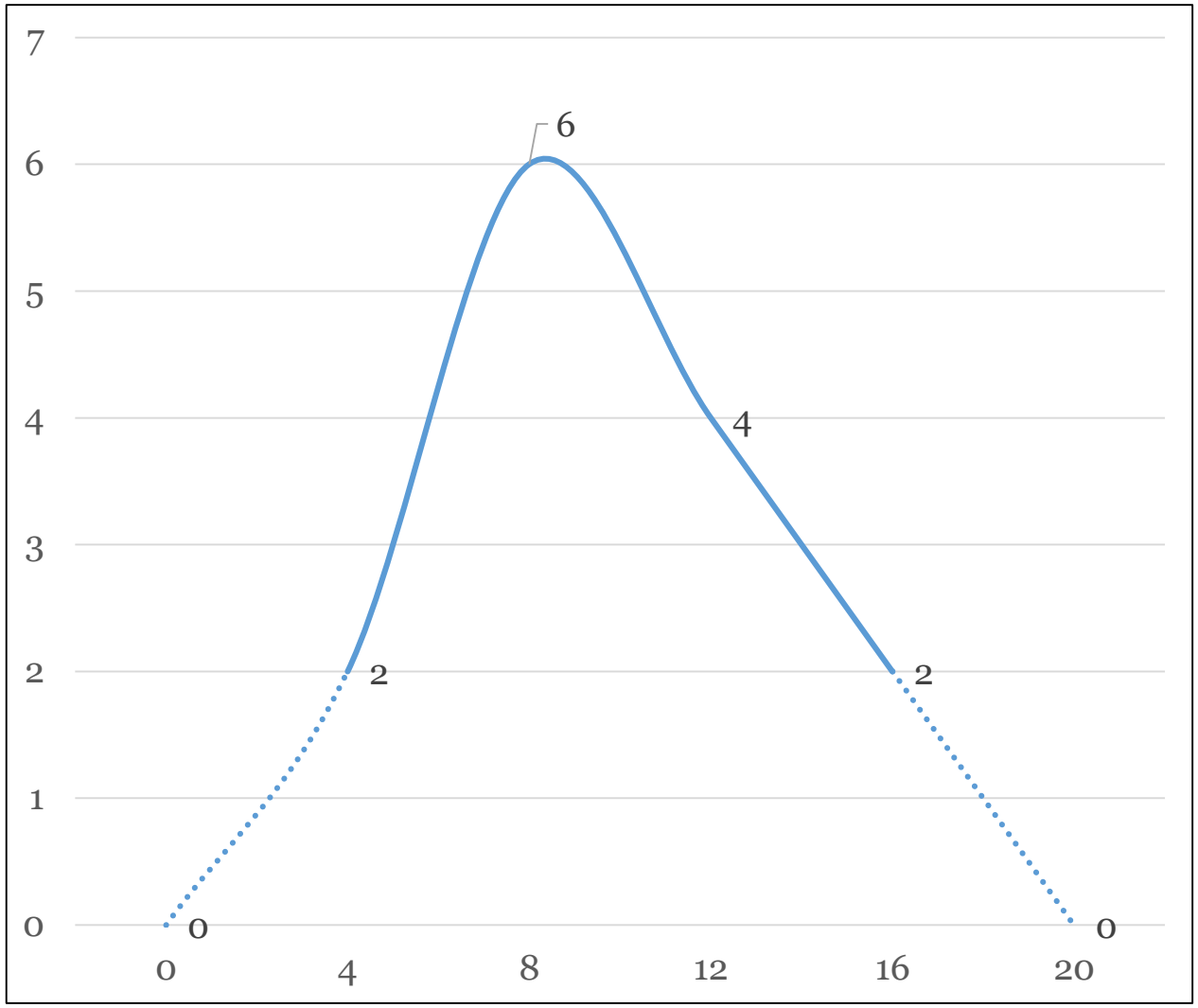
Frequency Polygon

Class	Mid point	Frequency
2-6	4	2
6-10	8	6
10-14	12	4
14-18	16	2



Frequency Curve

Class	Mid point	Frequency
2-6	4	2
6-10	8	6
10-14	12	4
14-18	16	2



Some points

- The histogram is more often used when single distributions are presented.

- The frequency polygon is largely used for comparison of two or more distribution.

Ogive curve

1. Firstly, we need to create a frame work (that is, X axis and Y axis)
2. Mark the distinct class boundary in X-axis
3. Mark the cumulative frequency in the Y-axis
4. Plot a point above each upper boundary (for less than type) at a height equal to the cumulative frequency of that class
5. Label the axis

Ogive Curve

- Distance of University from home (KM) for randomly selected students,

3, 8, 12, 9, 14, 6, 10, 11, 7, 5, 7, 16, 8,13

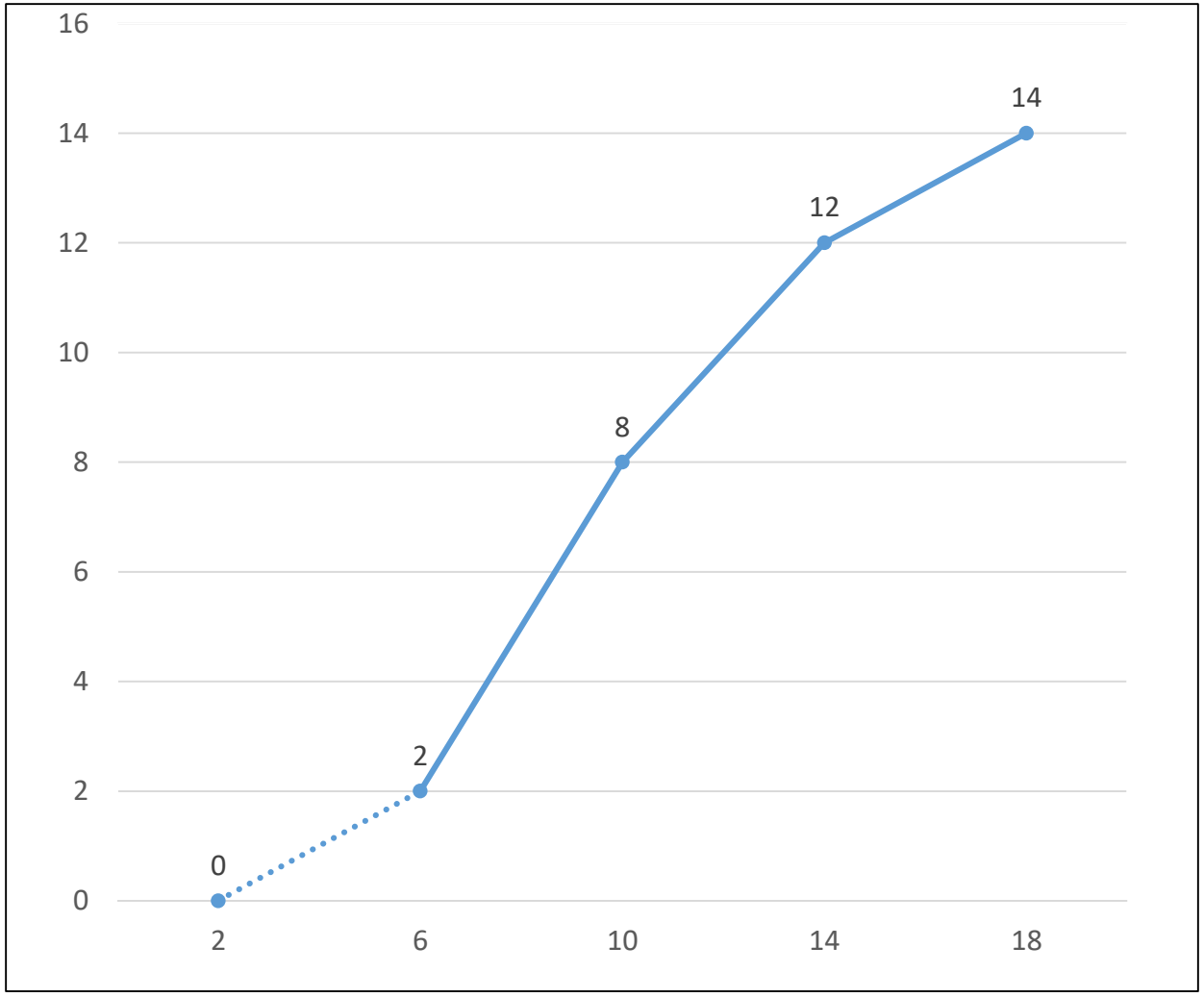
a) Construct the frequency distribution table.

b) Construct the ogive curve for the data.

Ogive Curve

Class	Frequency	Cumulative Frequency
2-6	2	2
6-10	6	8
10-14	4	12
14-18	2	14

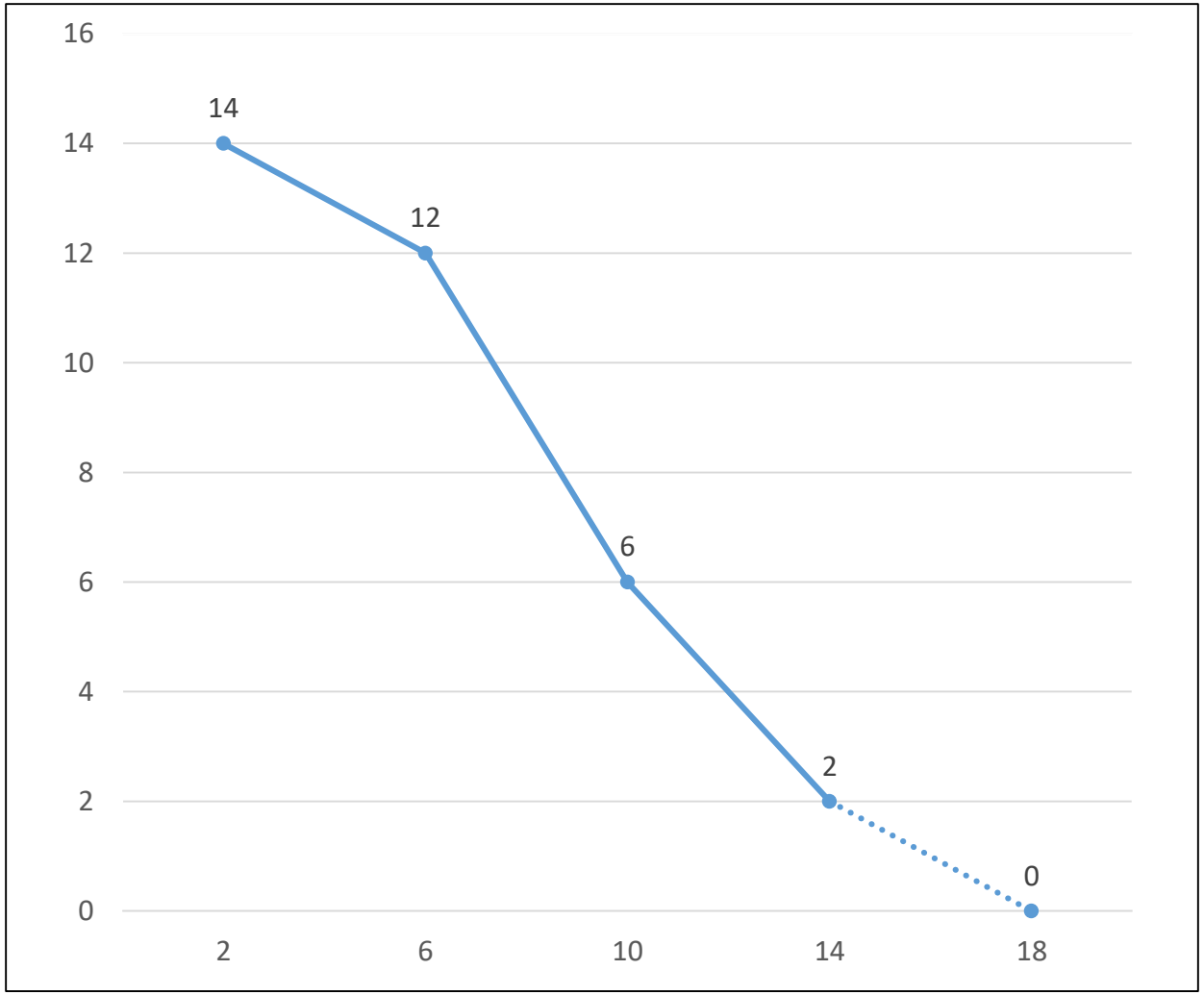
Less than type



Ogive Curve

More than type

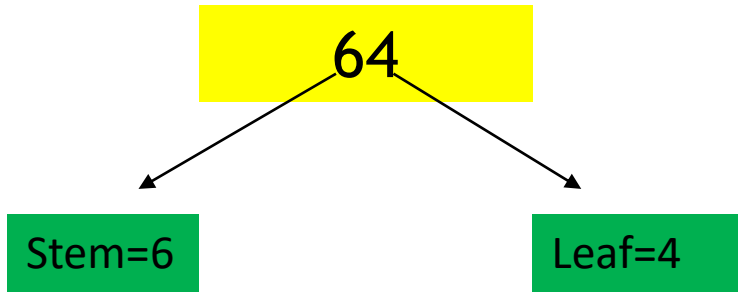
Class	Frequency	Cumulative Frequency
2-6	2	14
6-10	6	12
10-14	4	6
14-18	2	2



Stem and Leaf plot

- A stem-and-leaf display or stem-and-leaf plot is a device for presenting quantitative data in a graphical format.

- Where values of data are split into a stem and leaf. The first digit or digits will be written in stem and the last digit will be written in leaf.



Stem and Leaf

- Hypothetical data set:

3, 8, 12, 9, 14, 26, 10, 11, 7, 15, 7, 16, 8,13,22

a) Construct the stem and leaf for the data.

0	389778
1	2401563
2	62



0	377889
1	0123456
2	

Key: 1|2 represents 12

Bar chart vs Histogram

- Histogram refers to a graphical representation; that displays data by way of bars to show the frequency of numerical data.

- A bar graph is a pictorial representation of data that uses bars to compare different categories of data.

Important of graphs

1. Graphs are more attractive than tables of data
2. A graph gives a bird’s eye view of the entire data
3. Easy comparison among two or more data sets
4. Easy to interpret

Limitations of graphs

1. Amount of details is reduced

2. We cannot always compute precise estimates of the data sets

Thank You